



UITs

UNIVERSITY OF INFORMATION TECHNOLOGY & SCIENCES

Assignment

Course Code- CSE412

Course Title-Operating System Lab

<u>Submitted by-</u>	<u>Submitted to-</u>
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```

1  #include <iostream>
2  using namespace std;
3  struct Process {
4      int id, at, bt, ct, tat, wt;
5  };
6  int main() {
7      Process p[5];
8      int n = 5;
9      cout << "Enter Arrival Time and Burst Time for 5 processes:" << endl;
10     for (int i = 0; i < n; i++) {
11         p[i].id = i + 1;
12         cout << "Process " << p[i].id << " Arrival Time: ";
13         cin >> p[i].at;
14         cout << "Process " << p[i].id << " Burst Time: ";
15         cin >> p[i].bt;
16     }
17     for (int i = 0; i < n - 1; i++) {
18         for (int j = i + 1; j < n; j++) {
19             if (p[i].at > p[j].at) {
20                 swap(p[i], p[j]);
21             }
22         }
23     }
24     int time = 0;
25     for (int i = 0; i < n; i++) {
26         if (time < p[i].at) time = p[i].at;
27
28         p[i].ct = time + p[i].bt;
29         p[i].tat = p[i].ct - p[i].at;
30         p[i].wt = p[i].tat - p[i].bt;
31         time = p[i].ct;
32     }
33     cout << "\nProcess  AT\tBT\tCT\tTAT\tWT" << endl;
34     for (int i = 0; i < n; i++) {
35         cout << "P" << p[i].id << "\t" << p[i].at << "\t" << p[i].bt << "\t"
36             << p[i].ct << "\t" << p[i].tat << "\t" << p[i].wt << endl;
37     }
38     return 0;}

```

Enter Arrival Time and Burst Time for 5 processes:

Process 1 Arrival Time: 2
 Process 1 Burst Time: 3
 Process 2 Arrival Time: 4
 Process 2 Burst Time: 8
 Process 3 Arrival Time: 1
 Process 3 Burst Time: 6
 Process 4 Arrival Time: 4
 Process 4 Burst Time: 9
 Process 5 Arrival Time: 10
 Process 5 Burst Time: 11

Process	AT	BT	CT	TAT	WT
P3	1	6	7	6	0
P1	2	3	10	8	5
P2	4	8	18	14	6
P4	4	9	27	23	14
P5	10	11	38	28	17

=== Code Execution Successful ===